

Foundation (Jalapeno)

- Q1.** Rationalize the denominator of $\frac{2}{\sqrt{3}}$
- (A) $\frac{2\sqrt{3}}{3}$
(B) $\frac{2}{3\sqrt{3}}$
(C) $\frac{2\sqrt{2}}{3}$
(D) $\frac{\sqrt{2}}{3}$
(E) $\frac{2}{\sqrt{2}}$
- Q2.** A basketball player scored $\frac{1}{\sqrt{2}}$ points per minute. Rationalize the denominator to find the points scored in x minutes.
- (A) $\frac{x\sqrt{2}}{2}$
(B) $\frac{x}{2\sqrt{2}}$
(C) $\frac{x}{\sqrt{2}}$
(D) $\frac{x\sqrt{2}}{4}$
(E) $x\sqrt{2}$
- Q3.** The distance between two football players is $\frac{5}{\sqrt{11}}$ yards. Rationalize the denominator.
- (A) $\frac{5\sqrt{11}}{11}$
(B) $\frac{5}{11\sqrt{11}}$
(C) $\frac{55}{\sqrt{11}}$
(D) $\frac{5}{\sqrt{11}}$
(E) $5\sqrt{11}$
- Q4.** Rationalize the denominator of $\frac{3}{\sqrt{5+2}}$
- (A) $\frac{3(\sqrt{5}-2)}{5-4}$
(B) $\frac{3(\sqrt{5}+2)}{5+4}$
(C) $\frac{3\sqrt{5}}{5}$
(D) $\frac{3}{5\sqrt{5}}$
(E) $\frac{3(\sqrt{5}-2)}{5+4}$
- Q5.** The average score of a golfer is $\frac{72}{\sqrt{8}}$. Rationalize the denominator.
- (A) $\frac{72\sqrt{2}}{4}$
(B) $\frac{72}{4\sqrt{2}}$
(C) $\frac{72}{\sqrt{8}}$
(D) $\frac{36\sqrt{2}}{2}$
(E) $18\sqrt{2}$
- Q6.** The speed of a cyclist is $\frac{25}{\sqrt{10}}$ miles per hour. Rationalize the denominator.
- (A) $\frac{25\sqrt{10}}{10}$
(B) $\frac{25}{10\sqrt{10}}$
(C) $\frac{5\sqrt{10}}{2}$
(D) $\frac{25}{\sqrt{10}}$
(E) $5\sqrt{10}$
- Q7.** The time taken by an athlete to run x meters is $\frac{x}{\sqrt{6}}$ seconds. Rationalize the denominator.
- (A) $\frac{x\sqrt{6}}{6}$
(B) $\frac{x}{6\sqrt{6}}$
(C) $\frac{x}{\sqrt{6}}$
(D) $x\sqrt{6}$
(E) $\frac{x}{6}$
- Q8.** The distance traveled by a swimmer is $\frac{50}{\sqrt{3}}$ meters. Rationalize the denominator.
- (A) $\frac{50\sqrt{3}}{3}$
(B) $\frac{50}{3\sqrt{3}}$
(C) $\frac{50}{\sqrt{3}}$
(D) $50\sqrt{3}$
(E) $\frac{50}{3}$

Open Ended Questions (FRQ)

- Q9.** Rationalize the denominator of $\frac{2x+1}{\sqrt{x+2}}$ and simplify.

- Q10.** The score of a tennis player is $\frac{80}{\sqrt{16}}$. Rationalize the denominator and find the score.

Chef's Tips

- Check if the denominators are rationalized
- Verify the final answers
- Review the work for completeness and accuracy